

Multiple Choice (10 marks)

1. Find the GCD of 25 and 55.
 - (a) 275
 - (b) 15
 - (c) 20
 - (d) 50
 - (e) 11
2. Compute the area of the triangle whose altitudes have lengths 10, 12, and 15.
 - (a) $56\sqrt{15}$
 - (b) $60\sqrt{7}$
 - (c) $240\sqrt{7}/7$
 - (d) $48\sqrt{15}$
 - (e) $120\sqrt{15}$
3. A number x equals $2^{15} \cdot 3^6$. What number's cube equals x ?
 - (a) 125
 - (b) 343
 - (c) 64
 - (d) 216
 - (e) 512
4. The American flag is usually made with its width and length in the ratio of 10 to 19. Which dimensions are in the correct ratio for the flag?
 - (a) 27 ft by 53 ft
 - (b) 35 ft by 66 ft
 - (c) 25 ft by 47 ft
 - (d) 30 ft by 53 ft
 - (e) 30 in. by 57 in.
5. Let $I=[0,1]\times[0,1]$. Suppose $E = (x,y) \in I : \sin(x) < \frac{1}{2}, \cos(x+y) \text{ is irrational}$, what is the Lebesgue measure of E ?
 - (a) 0.7853981634
 - (b) 1.0471975512

- (c) 1.8849555922
- (d) 0.6283185307
- (e) 0.2617993833

True / False (10 marks)

Answer True or False

6. True or False: The correct answer to 'Is the Taylor Series for f at $x=5$ where $f(x)=\sum_{n=0}^{\infty} \frac{x^n}{n!} a_n$
6. True or False: The correct answer to 'Compute $\int_C dz/(z * (z - 2)^2)dz$, where $C: |z - 2| = 1$. The answer is Ai with i denoting the imaginary unit, what is A ?' is '1'.
7. True or False: Even with a standard deviation of zero, the sample can still have a skewed distribution due to the presence of outliers.
8. True or False: The function $f(x, y) = x^3 + y^3 + 3xy$ has local minima at both distinct points P and Q .
9. True or False: The probability that a random integer from the set S is divisible by 9 is 0.1005.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Considering a normally distributed waiting time for a new roller coaster ride with a mean of 35 minutes and a standard deviation of 10 minutes, analyze the time interval that encompasses the waiting times for 100,000 riders out of 150,000. Discuss the implications of this interval in the context of statistical distribution and rider experience.
12. Consider a point in space defined by coordinates (x, y, z) selected uniformly within the cube bounded by $-1 \leq x, y, z \leq 1$. Analyze the probability that this point lies within the unit sphere defined by the equation $x^2 + y^2 + z^2 \leq 1$. Discuss the geometric interpretation and the mathematical principles involved in your analysis.

End of Paper